Léo Martin

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Computer science engineering student specializing in real-time computer vision (object detection, multi-object tracking, and edge deployment), validated through rigorous quantitative evaluation. Currently an AI apprentice engineer at Thales on real-time maritime perception. Seeking an international research internship on real-time perception, physical AI, or human-robot interaction. Available June–September 2027 (~17 weeks).

Skills

AI / Computer Vision: Python, PyTorch, TensorRT, OpenCV, Ultralytics YOLO, object detection & tracking (RF-DETR, RT-DETR), CVAT

Languages: Python, C++, C, C#, JavaScript, PHP, SQL

Web & Frameworks: React, Next.js, Node.js/Express, .NET

Databases: PostgreSQL, MongoDB, SQL

DevOps / Tools: Docker, Linux, Git/GitHub/GitLab

Education

CESI Rouen — Rouen, France
Engineering Degree, Computer Science (apprenticeship) — *2025 – 2028*
Project-based curriculum: information systems & security, software engineering, operations research, distributed applications

CESI Rouen — Rouen, France
Integrated Preparatory Cycle — *2023 – 2025*
Embedded systems, object-oriented programming, networks, databases, web development

Lycée Guillaume Le Conquérant — Lillebonne, France
French Baccalauréat, Science track (with honours) — *2021*
Majors: Mathematics, Life Sciences

Experience

Thales LAS ACR — Ymare, France
Apprentice Engineer – AI Development & Data Analysis — *Oct 2025 – Present*
AI-driven video turret proof of concept:

- Detect and track maritime targets in real time, with closed-loop PTZ camera control to keep the target in frame
- Train and benchmark two recent detectors (RF-DETR, RT-DETR) on an in-house annotated maritime dataset
- Optimise inference for real-time operation using TensorRT and deploy on an air-gapped server
- Build an active learning loop where live operator corrections feed model retraining

Data Analyst Intern — *Apr 2025 – Jul 2025*

- Managed and restructured the surveillance radar database, improving data reliability and query efficiency
- Maintained Excel tracking tools and developed data validation tooling
- Streamlined the collection and integration of data from multiple sources

Projects

Padel Analysis - Monocular sports perception & event reconstruction — Python · PyTorch · Ultralytics YOLO · OpenCV

- Built an end-to-end pipeline detecting and tracking players and ball from a single camera: players (0.88 F1, 0.84 IDF1), ball (80% within 10 px) on unseen footage
- Classified ball-surface contacts (racket, floor, glass, net) with a temporal network combining trajectory, motion and court geometry, reaching 82–83% accuracy on unseen minutes and 86% on an unseen tournament
- Reconstructed 3D contact positions from the single camera using court geometry, inferring occluded glass contacts from ball physics

Breezy - Distributed lightweight social network — Node/Express · React/Next.js · JWT · MongoDB/PostgreSQL · Docker

- Microservices-based social network with independent services communicating over a shared architecture, containerised with Docker

EasySave - Backup software — C# · .NET

- Backup software over 3 releases, evolving from a console app to an MVVM architecture with parallel multi-threaded transfers and encryption

TSP Resolver - Operations research — Python

- Complexity analysis of a constrained TSP-type routing problem, with metaheuristics implemented to approximate solutions

Languages & Interests

Languages: French (native), English (B2), Spanish (B1)

Interests: Sport (football, weight training, basketball), Travel, Reading, Music (piano)